

Sampling Theory

Design vs Model-based Sampling, Bias–variance, bullseyes and spatial sampling in R

Dr K.J. Mhango

Today's session

- **Part 1 – Bias & variance**
 - What do we mean by *bias* and *variance*?
 - Intuition via a **bullseye** (target) analogy
 - Why both matter for sampling and estimation
- **Part 2 – Sampling theory**
 - **Design-based** vs **model-based** inference
 - Typical sampling designs: SRS, stratified, cluster
- **Part 3 – Spatial sampling in R**
 - Build a simple “study region”
 - Implement convenience, random, stratified, and cluster sampling
 - Visualise what our samples “look like” in space

Part 1 – Bias & variance

- Intuition via a **bullseye** metaphor
 - How bias and variance trade off
 - Why both matter for sampling and estimation
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Bias vs variance – intuition

- Imagine we are **throwing darts** at a bullseye.
- The **true value** is the centre of the target.
- Each dart is an **estimate** (e.g. mean from a sample).

We care about **two things**:

- **Bias**: Are our darts **systematically off-centre**?
- **Variance**: Are our darts **tightly clustered** or **widely spread**?

Ideally: **low bias & low variance**

→ darts tightly clustered **around the centre**.

Bullseye scenarios

Bias–variance bullseye scenarios

Different combinations of bias and variance

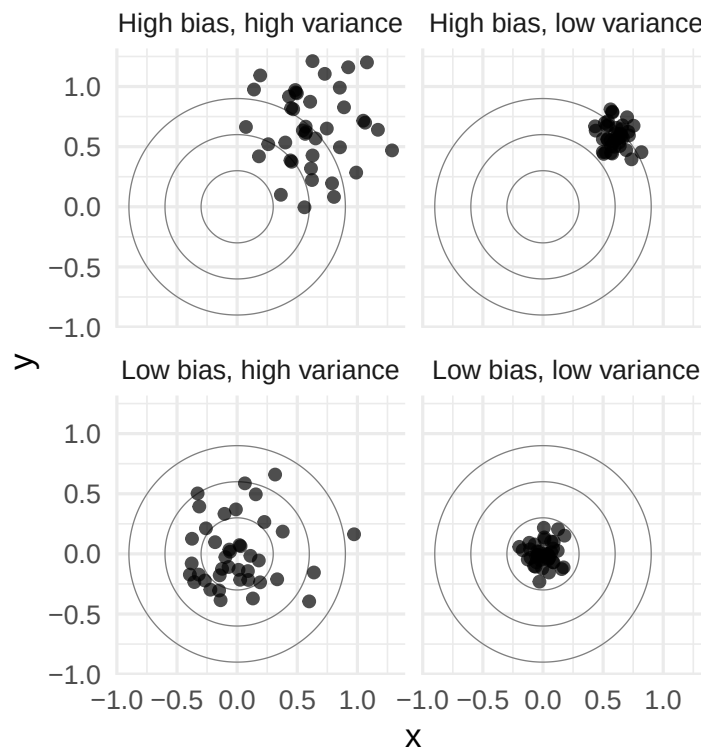


Figure 1: Bias–variance bullseye plots

Part 2 – Sampling theory

- How inference depends on your **sampling design**
 - **Design-based vs model-based** thinking
 - Error, power, and effect size
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Design-based vs model-based inference

- **Design-based inference:** randomness comes from the *sampling design* (e.g., SRS, stratified, cluster).
Estimators are justified by randomization; validity does not rely on a strong model for the population.
 - **Model-based inference:** treat the population as a realization of a **statistical model** (e.g., spatial regression, Gaussian field).
Inference is conditional on the model being approximately correct.
 - In practice, **design-based approaches are robust but can be inefficient**, while **model-based approaches can be very efficient but fragile if the model is wrong**.
 - Many modern surveys use a **hybrid strategy**: a good probability design plus a model to improve precision.
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Typical sampling designs (conceptual)

- **Simple random sampling (SRS):** every unit has equal chance; conceptually simple, design-based gold standard.
 - **Stratified sampling:** partition the population into strata and sample within each; improves precision and ensures coverage.
 - **Cluster sampling:** sample whole groups (clusters) of nearby units; cheaper logistically but with larger sampling error.
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How do we decide on a sample size?

Determining **how many samples** you need is as fundamental as choosing *where* they should come from. In environmental science, sample size influences cost, effort, accuracy, and the credibility of your conclusions.

Sample size is driven by four quantities:

- **Effect size:** What is the smallest meaningful difference or pattern you want to detect?
 - **Variance:** How noisy is the underlying process? More variance $\rightarrow$ larger n .
 - **Significance level (α):** Probability of a **Type I error** (false positive).
 - **Power ($1 - \beta$):** Probability of correctly detecting a true effect.
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Type I and Type II errors

Type I error (α)

A **false positive** — concluding an effect exists when it does *not*. We usually choose $\alpha = 0.05$.

Type II error (β)

A **false negative** — failing to detect a real effect. $\text{Power} = 1 - \beta$ measures how likely we are to detect the effect.

Typical scientific studies aim for **80% power**, meaning $\beta = 0.20$.

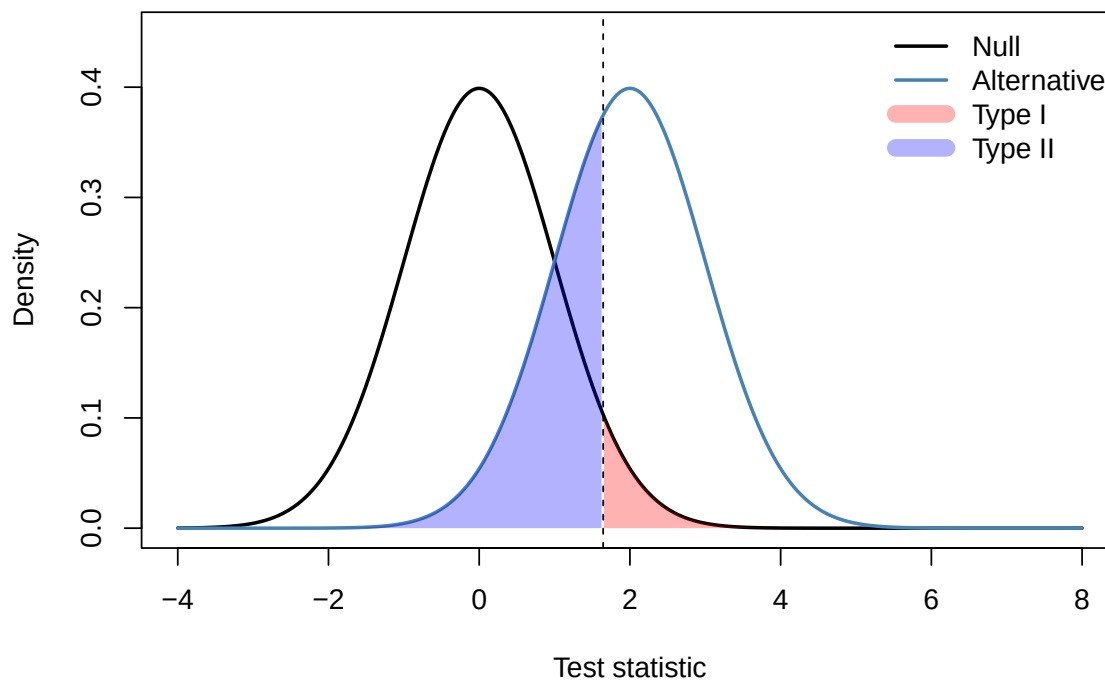


Figure 2: Type I and II error regions

Example: Power analysis in R

Below is an example power calculation for a **two-sample t-test** when the effect size varies.

```
library(pwr)

effects <- seq(0.2, 1, by = 0.05)
res <- sapply(effects, function(d) pwr.t.test(d = d, power = 0.8, sig.level = 0.05, type = "t"))

plot(effects, res, type = "l", lwd = 2,
     xlab = "Effect size (Cohen's d)",
     ylab = "Required sample size per group",
     main = "Smaller effects require larger samples")
```

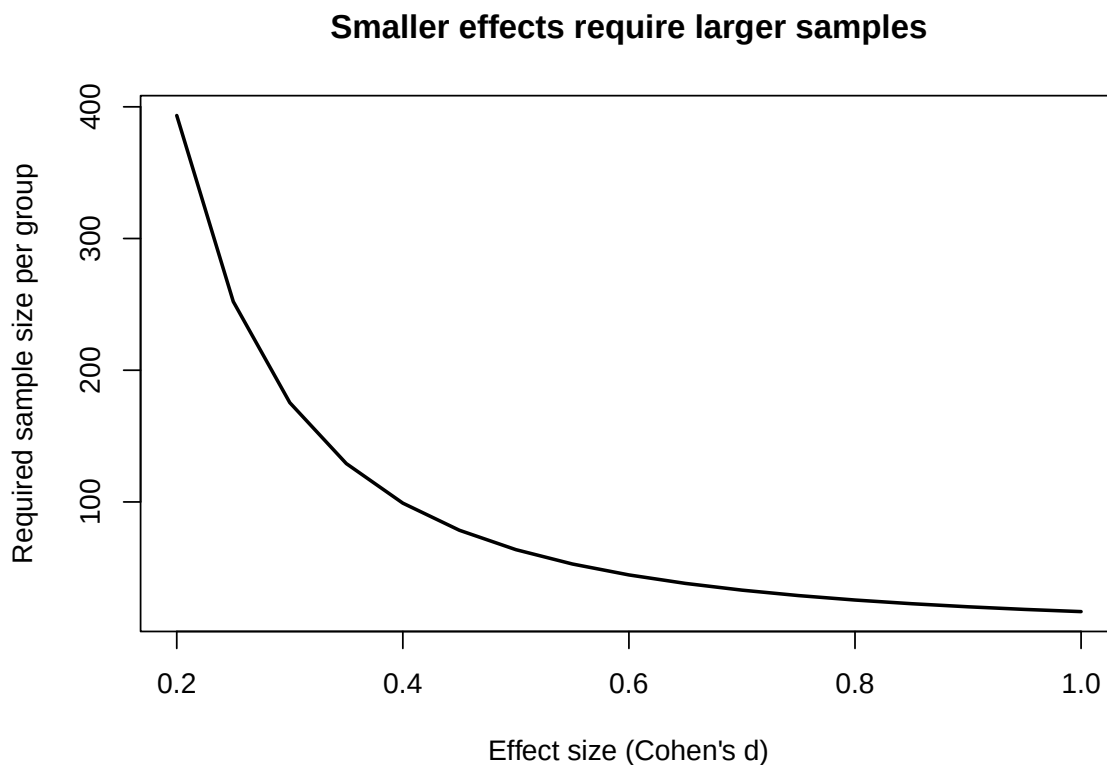


Figure 3: Required sample size as a function of effect size

Practical guidance

- **Effect size thinking matters:** trivial differences do not justify massive sampling campaigns.
 - **More variance → more samples:** noisy ecological processes require deeper sampling.
 - **Design-based sampling controls bias;** model-based approaches may be more efficient *if the model is correct*.
 - **Alpha and power trade-offs** must be justified, not chosen arbitrarily.
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Summary so far – theory

We now have: - A conceptual grounding in **bias and variance** - A working picture of **design-based vs model-based** inference - An overview of common designs: **SRS, stratified, cluster** - Theoretical tools for **sample size, effect size**, and **Type I / II errors**.

Next we'll see how these ideas play out in **spatial sampling in R**.

Part 3 – Spatial sampling in R

- Build a simple grid-based **study region**
 - Compare **convenience, random, stratified, and cluster** designs
 - Visualise how design affects **spatial coverage and bias**
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Spatial example – create a study region in R

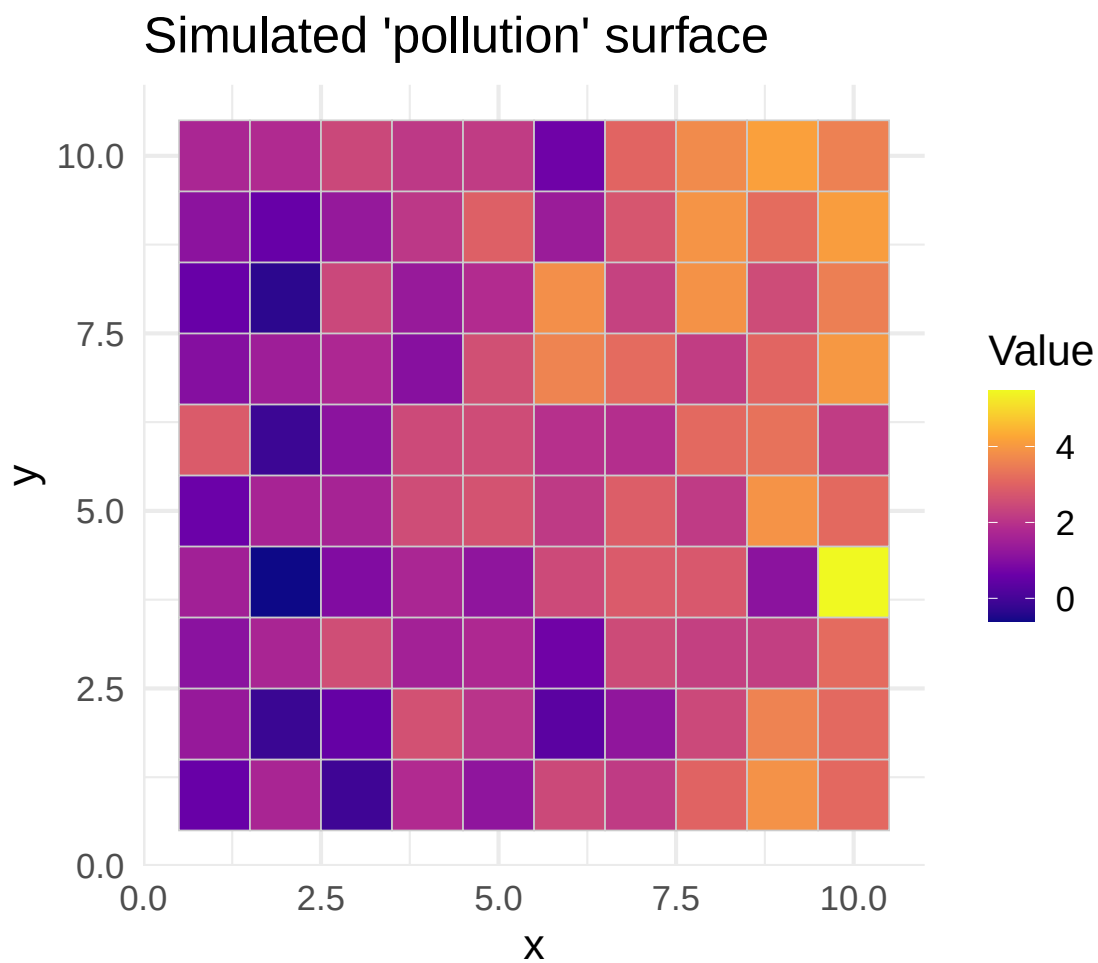


Figure 4: Simulated study region (grid)

Convenience sampling

- **Convenience sampling** selects units that are easiest to reach or access, rather than using random or systematic selection.
- This approach is often quick and low-cost, relying on availability rather than a defined sampling plan.
- It does not ensure that every member of the population has a chance of selection, introducing **selection bias**.
- As a result, the sample may not be representative, and findings may not generalize to the broader population.
- In environmental studies, examples include sampling near roads, field stations, or other easily accessible sites.

Convenience sampling: corner only

High risk of bias

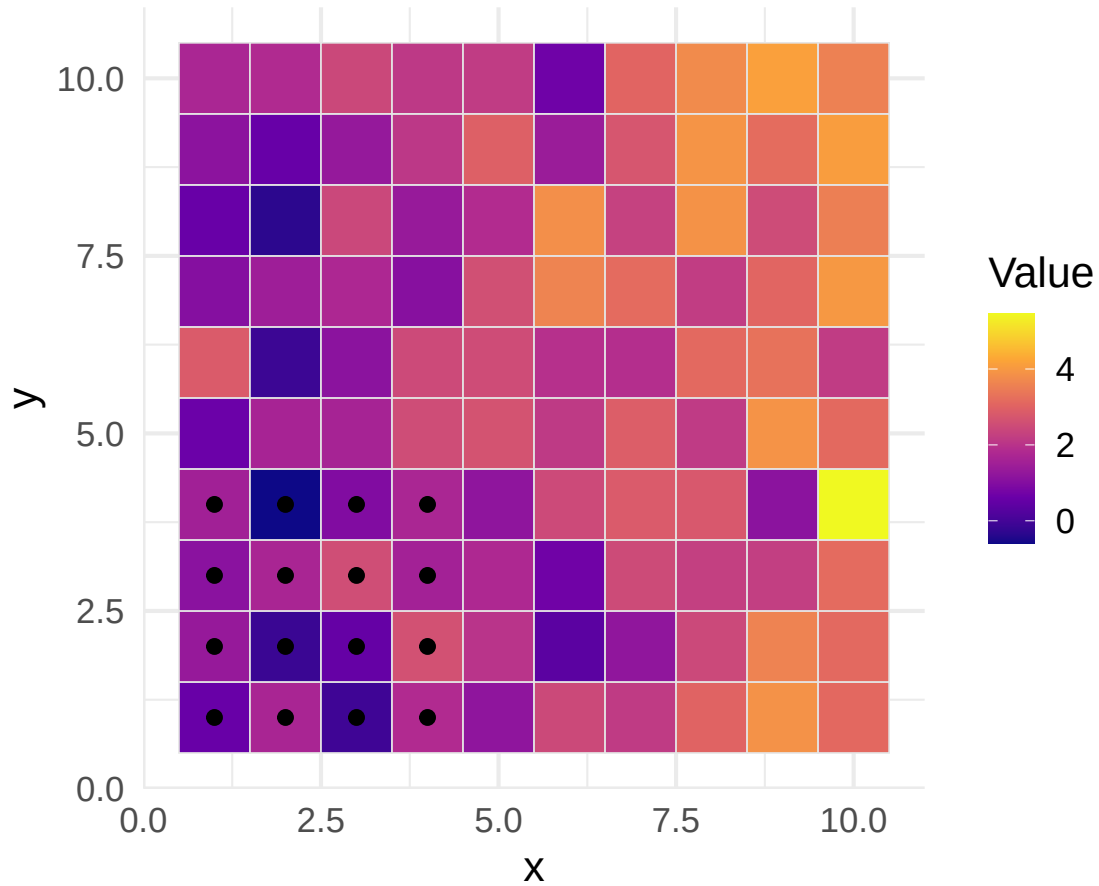


Figure 5: Convenience sample in the bottom-left area

Simple random sampling (SRS)

- **Simple Random Sampling (SRS)** is a method where every unit in the population has an equal and independent chance of being selected.

- SRS does not consider spatial, temporal, or group structure—samples are chosen uniformly from the entire sampling frame.
- This approach is easy to implement and provides unbiased estimates as long as all units are accessible.
- It may be inefficient in populations with strong spatial structure or gradients, potentially missing some areas entirely.
- SRS requires a complete list of all units and may not be practical when access or sampling is logistically constrained.

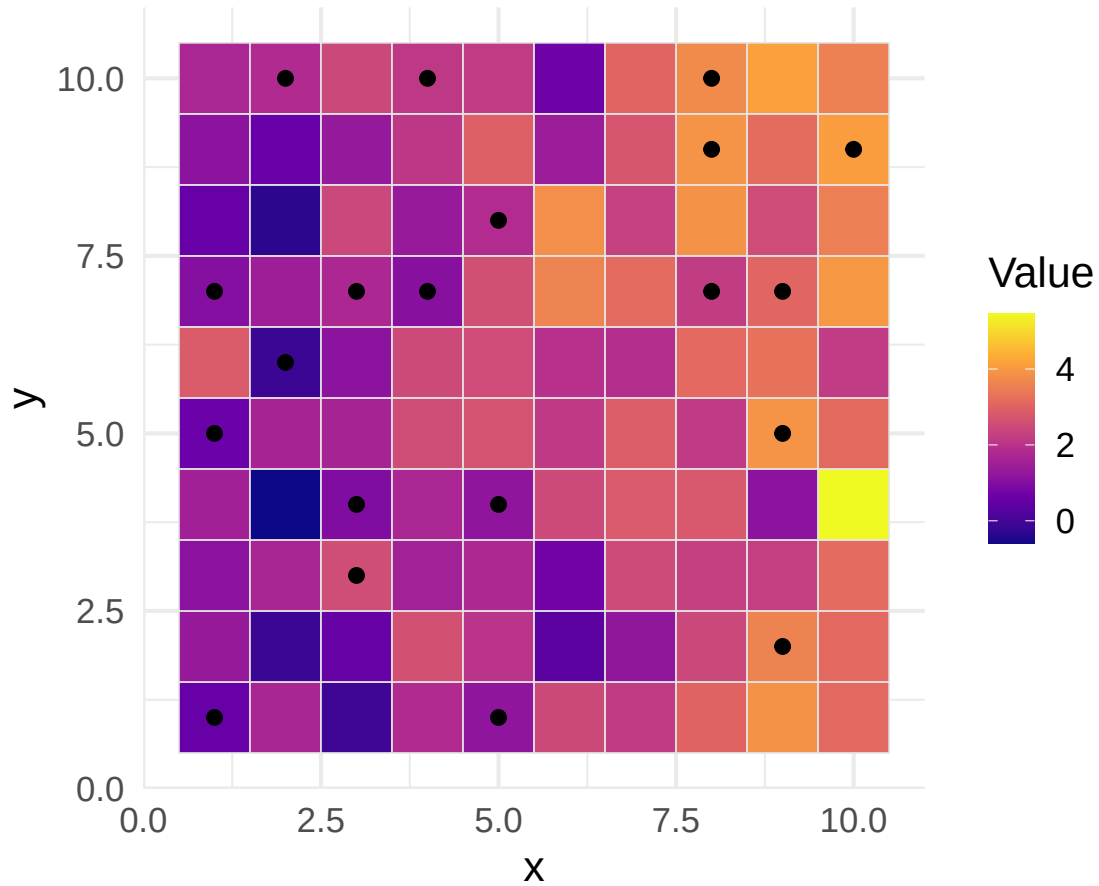
```
n_srs <- 20

srs_sample <- grid |>
  sample_n(n_srs)

ggplot() +
  geom_tile(data = grid, aes(x, y, fill = value), color = "grey90") +
  geom_point(data = srs_sample, aes(x, y), color = "black", size = 2) +
  scale_fill_viridis_c(option = "C") +
  coord_equal() +
  labs(title = paste0("Simple random sample (n = ", n_srs, ")"), subtitle = "Unbiased under SRS")
```

Simple random sample (n = 20)

Unbiased under SRS



Stratified sampling

- **Stratified sampling** divides the population into distinct subgroups (*strata*) based on important characteristics (such as habitat, region, or demographic group).

- **Samples are randomly selected from within each stratum**, ensuring that all segments of the population are represented in the final sample.
- **Improves precision** of estimates, especially when there is substantial difference between strata but more similarity within each stratum.
- The number of samples from each stratum can be **proportional to its size** or allocated equally, depending on study goals.
- **Analysis must account for stratification** by appropriately weighting or aggregating results across strata to produce unbiased population-level estimates.

```
grid_strat <- grid |>
  mutate(
    stratum = cut(
      y,
      breaks = c(0, 3.33, 6.66, 10.1),
      labels = c("South", "Middle", "North")
    )
  )

n_per_stratum <- 8

strat_sample <- grid_strat |>
  group_by(stratum) |>
  sample_n(n_per_stratum) |>
  ungroup()

ggplot() +
  geom_tile(data = grid_strat, aes(x, y, fill = value), color = "grey90") +
  geom_point(data = strat_sample, aes(x, y, shape = stratum), color = "black", size = 2) +
  scale_fill_viridis_c(option = "C") +
  coord_equal() +
  labs(title = "Stratified sampling", subtitle = "Better coverage across landscape", fill = "value")
```

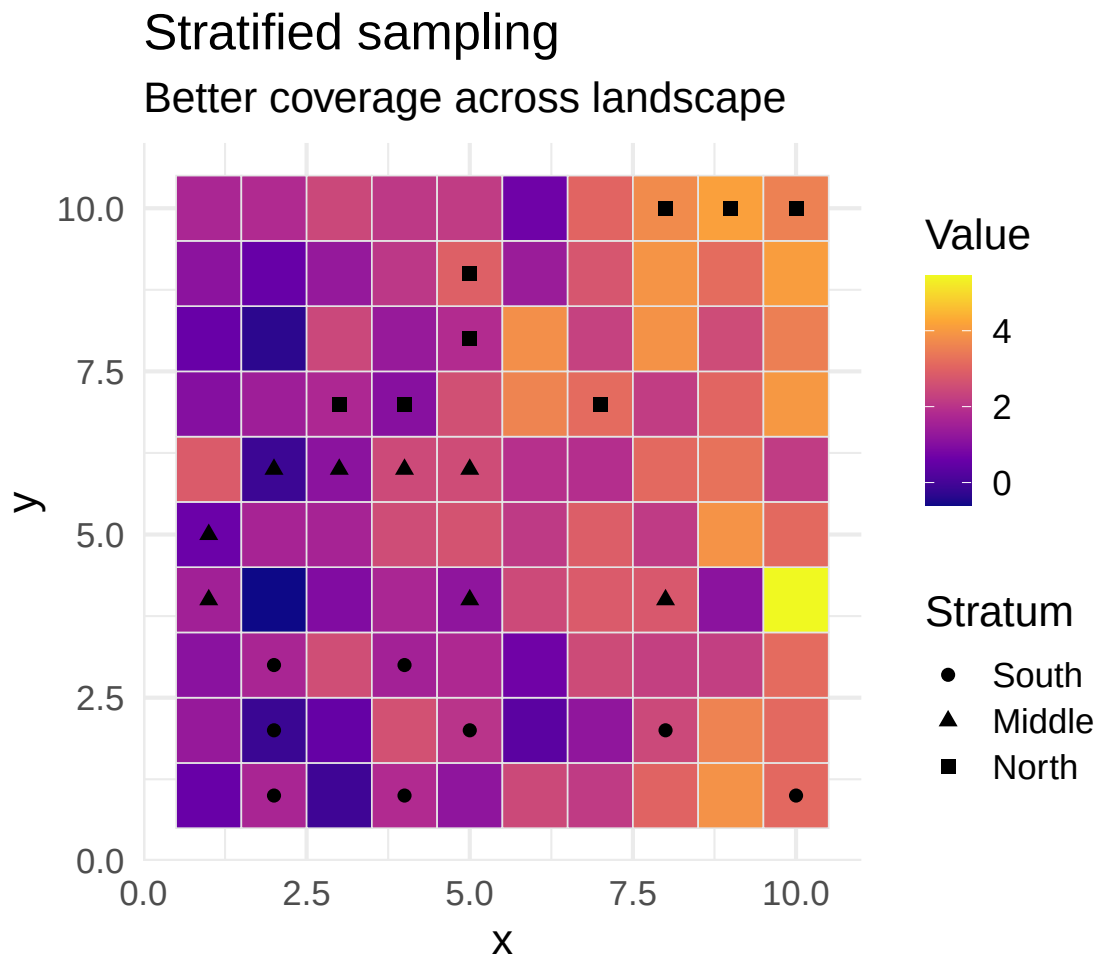


Figure 7: Stratified sampling by north/south bands

Cluster sampling

- **Cluster sampling** groups the population into clusters (e.g., spatial blocks, villages, or schools) that are often based on natural or convenient divisions.

- **Randomly selects entire clusters**, then samples all (or a subset of) units within those chosen clusters, rather than drawing from the population at large.
- **Reduces cost and logistical effort**, since sampling is concentrated in a small number of locations, making it practical for widely dispersed populations.
- **Works best when clusters are similar to each other but internally heterogeneous**, so that each selected cluster is representative of the whole population.
- **Statistical analysis must account for the cluster structure**, as observations within clusters may be more alike—this increases sampling error compared to simple random sampling.

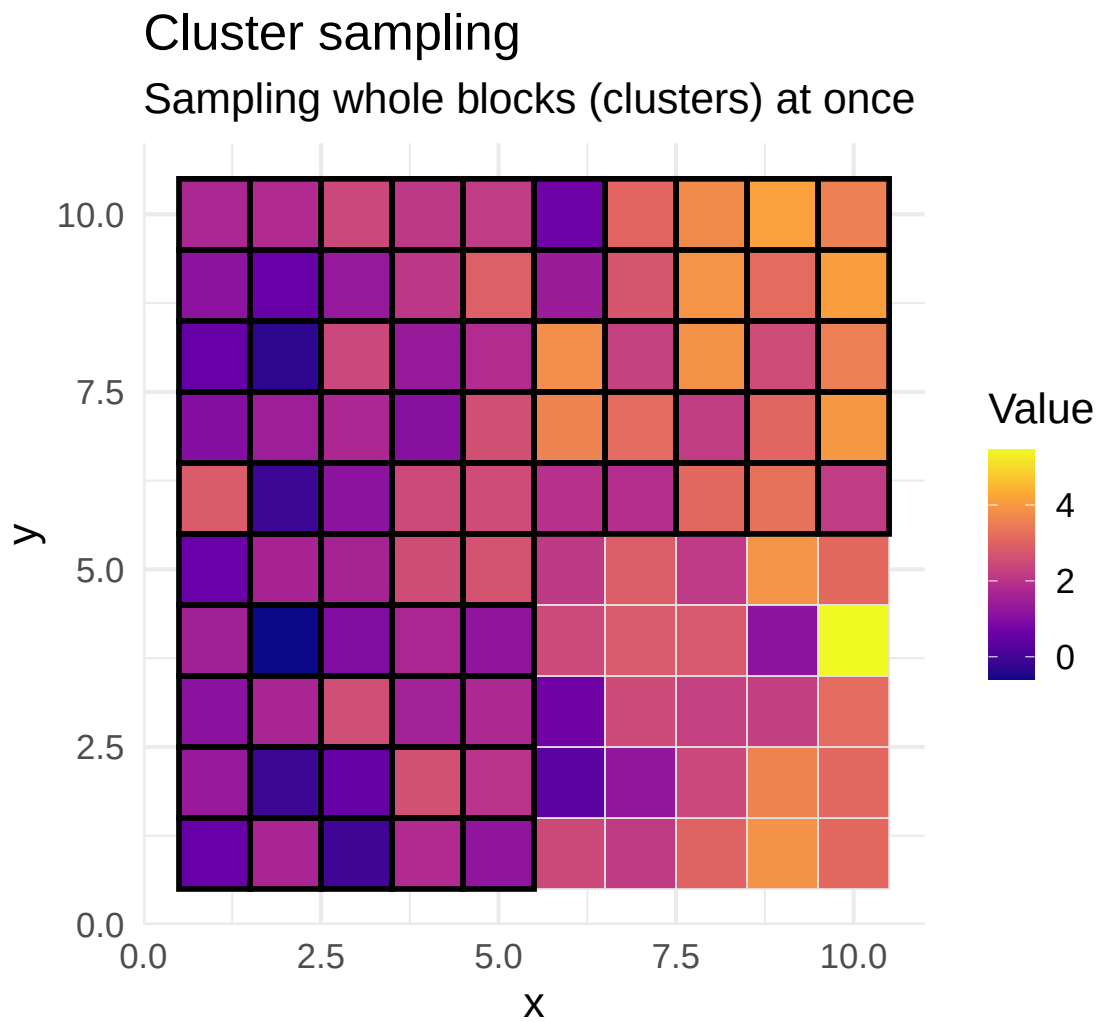


Figure 8: Cluster sampling using 5×5 blocks

How are clusters and strata created?

- **Clusters** and **strata** are groupings of population units, but they differ in purpose and how they are created.

- **Clusters:** Groups of units *sampled together*—often to reduce cost/logistics.
- **Strata:** Groups of units that are *internally similar*, used to ensure groups are **represented** in the sample.

The *definition* of clusters or strata can greatly impact the results of sampling—so, how do we create them?

Judgemental (Subjective) Approaches

- **Expert knowledge:** Clusters or strata are defined by people who know the study area well (e.g., separating neighbourhoods by known boundaries, land use, or socio-economic status).
 - **Administrative boundaries:** Using counties, postal codes, schools, etc.
 - **Pro:** Fast and accessible; uses existing groupings.
 - **Con:** Can introduce bias, as groupings may not reflect underlying variation.
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Model-based & Data-driven Approaches

- **Use of auxiliary data** (e.g., satellite data, previous surveys, environmental layers) to identify groups that are similar or different.
 - **Spatial models:** Delineate regions based on predicted values from spatial models (e.g., soil properties, health metrics).
 - **Pro:** Can tailor clusters/strata to variation in the variable of interest.
 - **Con:** Requires sufficient, relevant data and analysis.
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Statistical Methods – Example: k-means Clustering

- **k-means** is an algorithm that partitions units into (k) clusters so that units within a cluster are more similar to each other than to those in other clusters.
- **Steps:**
 1. Specify number of clusters ((k)).
 2. Randomly assign initial cluster centers.
 3. Assign each unit to the nearest center.

4. Update centers and repeat until stable.
- **Application:** Often used to create clusters or strata using continuous auxiliary variables (e.g., income, environmental gradients).

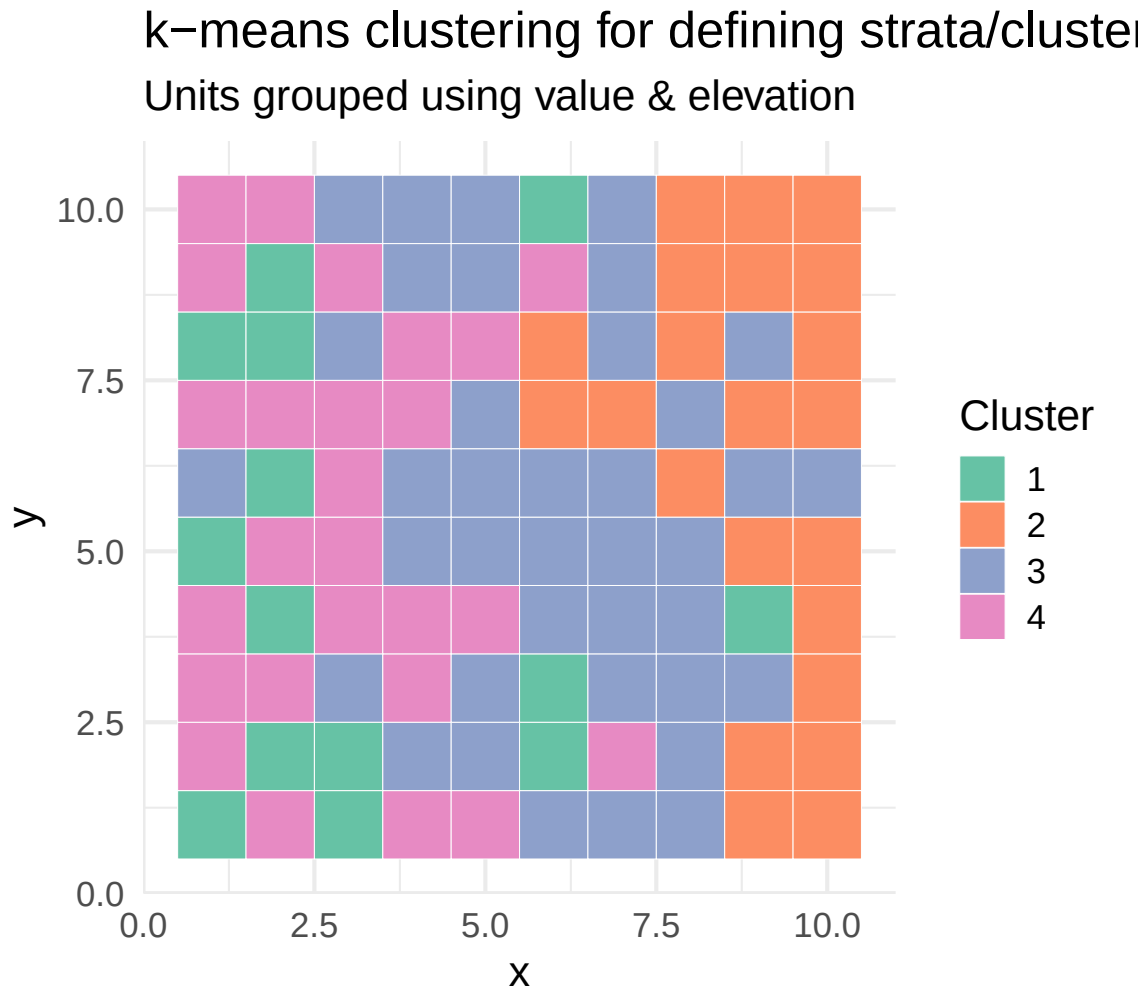


Figure 9: Creating strata/clusters using k-means clustering on two variables

Comparison: Choosing a Method

- **Judgemental** methods: Simple, transparent, but can miss important variation.
- **Model-based** groupings: Potentially more representative, but require good covariate data.
- **Statistical clustering (e.g., k-means)**: Objective, reproducible, works for multi-variate data, but needs careful selection of variables.

The best approach depends on your study's goals, available data, and logistical constraints.
